

Do Architecture Families Leave Fingerprints in Adversarial Failures? Measuring Δ^* -Vector Profiles Across Transformer Families

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Abstract

Can a language model’s architecture family be identified from its adversarial failures alone? Current robustness assessments treat models as individuals, aggregating vulnerability into a scalar and discarding the multivariate structure across attack types that architecture constrains. This paper introduces the Δ^* -vector framework, which represents each model as a profile of normalized per-attack-category vulnerability scores and tests whether architecture family (encoder-only, decoder-only, encoder-decoder) explains this profile. Applied to nine scale-matched transformer models across six adversarial attack categories (AdvGLUE and ANLI-R3), permutation MANOVA yields $\eta^2 = 0.293$ — a large effect (Cohen’s $f^2 \approx 0.41$) that exceeds the pre-specified $\eta^2 = 0.15$ threshold in five of six attack categories (83%). Statistical significance at $\alpha = 0.05$ is not achieved ($p=0.147$, 1,000 permutations), consistent with the approximately 40% statistical power available at $N=9$; $N \geq 15$ (≥ 5 per family) is required for 80% power. Leave-one-model-out (LOMO) classification achieves accuracy=0.333 — at three-class chance — a result attributable to geometric degeneracy at $N=3$ models per family rather than absence of family structure. A validated seven-module pipeline (1,788 lines, Python) is provided. The existence signal is confirmed at effect-size level; overall statistical significance awaits a confirmatory study with an expanded model pool.

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1. Introduction

Can a language model’s architecture family be identified from its adversarial failures alone — without knowing its weights, training data, or any other meta-data? This paper investigates whether that answer is yes, at least in principle. Architecture family membership (encoder-only, decoder-only, encoder-decoder) accounts for approximately 29% of variance in normalized adversarial vulnerability across attack categories at base model scale (permutation MANOVA $\eta^2 = 0.293$, Cohen’s $f^2 \approx 0.41$), and this effect exceeds a pre-specified practical threshold in 83% of the adversarial attack categories evaluated. The signal is large. However, statistical confirmation requires a larger model pool than the nine models studied here — a limitation this paper quantifies as a concrete power analysis and study-design contribution.

The practical context motivates the question. A security auditor comparing encoder-only models (e.g., BERT-style) against decoder-only models (e.g., GPT-style) for adversarially sensitive deployment cannot currently quantify their different adversarial risk profiles from first principles. Robustness assessments today are model-specific and non-transferable: each new model requires a fresh evaluation from scratch. Architecture-family-level fingerprinting, if empirically confirmed, would allow principled inference about an unseen model’s likely vulnerability profile from its architecture family membership alone, reducing evaluation cost substantially.

A deeper structural problem underlies evaluation cost. Existing adversarial robustness studies evaluate models as individuals, reporting scalar accuracy-drop metrics aggregated across attack types. Scalar aggregation discards the multivariate structure of adversarial vulnerability. Different attack types stress different model components: whether a model is more vulnerable to lexical substitutions than to paraphrase attacks may be systematically related to how its attention mechanism processes input perturbations. Encoder-only models with bidirectional attention globally redistribute attention mass over the full input sequence

under perturbation; decoder-only models with causal attention process perturbations locally in sequence order. These structural differences, the present study hypothesizes, produce characteristic per-attack-category vulnerability profiles — a Δ^* -vector — that are consistent within architecture families and discriminable between them.

Prior work provides directional evidence. Neerudu et al. (2023) compare BERT, GPT-2, and T5 on GLUE perturbations and find GPT-2 (decoder-only) more robust than BERT (encoder-only) and T5 (encoder-decoder), a result consistent with architecture-family effects. Wang et al. (2021) establish AdvGLUE as a multi-task adversarial benchmark. Nie et al. (2020) provide ANLI as a human-crafted adversarial NLI benchmark. Zhang et al. (2026) find lower explanation flip rates for decoder LLMs in enterprise NLP contexts. Yet none of these works formalizes the Δ^* -vector representation, quantifies between-family effect size, or applies multivariate testing suited to small model pools. Sun et al. (2024, TrustLLM) evaluate 16 LLMs across trustworthiness dimensions but do not stratify by architecture family. Otmakhova et al. (2025, FLUKE) demonstrate that linguistic capability does not imply robustness to perturbations of that feature — motivation consistent with the present approach but orthogonal in method.

This paper makes four contributions. First, it introduces the Δ^* -vector framework **for multivariate architecture-family adversarial profiling: each model is represented as a vector of normalized per-category vulnerability scores, and architecture-family effects are tested with permutation MANOVA. Second, it provides the first formal effect-size measurement* of between-family Δ^* -vector clustering: $\eta^2 = 0.293$ across nine models and six attack categories, exceeding the pre-specified $\eta^2 = 0.15$ threshold in five of six categories. Third, it contributes a **validated seven-module pipeline** (1,788 lines) covering fine-tuning via pre-existing checkpoints, adversarial evaluation, Δ^* computation, statistical analysis, and visualization. Fourth, it provides a **power analysis** establishing $N \geq 15$ (≥ 5 models per family) as the minimum for 80% statistical power at the observed effect size — concrete design guidance for a confirmatory follow-up study.**

The paper proceeds as follows. Section 2 reviews adversarial robustness benchmarks, architecture comparison studies, and LLM trustworthiness evaluation. Section 3 describes the Δ^* -vector framework and statistical methods. Section 4 presents the experimental setup. Section 5 reports results. Section 6 discusses implica-

tions and limitations. Section 7 concludes.

2. Related Work

2.0.1. 2.1 ADVERSARIAL ROBUSTNESS BENCHMARKS

Ribeiro et al. (2020) introduced CheckList, a behavioral testing paradigm evaluating NLP models across a matrix of linguistic capabilities and perturbation types. CheckList shifts the evaluation frame from aggregate accuracy to structured failure-mode analysis, but does not compare architecture families. Wang et al. (2021) released AdvGLUE, a multi-task adversarial benchmark derived from five GLUE tasks using 14 attack methods spanning character-level noise, word-level substitution, and semantic perturbations; it is the primary evaluation suite in this study. Nie et al. (2020) developed ANLI (Adversarial NLI), where examples are collected via iterative human-model adversarial interaction, providing a benchmark resistant to statistical artifacts from automatic perturbation methods. ANLI Round 3 (ANLI-R3, 1,200 examples) is used here as the human-crafted, surrogate-free evaluation partition.

A persistent limitation in prior use of these benchmarks is their application for leaderboard-style scalar comparison. The present work instead treats AdvGLUE and ANLI-R3 as sources of per-category Δ^* observations and subjects the resulting attack-category $\times$ model matrix to multivariate analysis.

2.0.2. 2.2 ARCHITECTURE COMPARISON STUDIES

Neerudu et al. (2023) compare BERT, GPT-2, and T5 on GLUE with 8 text perturbation types and find that GPT-2 representations are more robust than BERT and T5 across multiple perturbation categories. This directional finding is consistent with the $\eta^2 = 0.293$ result reported here and with the attention-topology hypothesis. However, Neerudu et al. evaluate only three models (one per family), report no effect size, apply no multivariate test, and do not use AdvGLUE or ANLI. The present work extends that directional observation with formal effect-size quantification, a nine-model pool, a six-category Δ^* -vector representation, and permutation MANOVA with statistical controls.

Zhang et al. (2026) find that decoder-only models show lower explanation flip rates than encoder baselines in enterprise NLP settings, consistent with architecture-family robustness differences in deployment contexts. Yang et al. (2024) compare LLM robustness under white-box attacks and observe that model size and structure affect robustness, without for-

mal between-family effect-size quantification.

2.0.3. 2.3 LLM TRUSTWORTHINESS EVALUATION

Sun et al. (2024, TrustLLM) evaluate 16 LLMs across six trustworthiness dimensions using more than 30 datasets. The robustness dimension is relevant to the present study, but TrustLLM does not stratify by architecture family or compute per-attack-category Δ -vectors. Otmakhova et al. (2025, FLUKE) demonstrate that a model’s ability to use a linguistic feature does not imply robustness to perturbations of that feature, motivation consistent with the present approach of measuring Δ independently of clean accuracy. Cohen (1988) provides the multivariate effect-size conventions and power analysis framework applied throughout this paper.

Summary. Prior work establishes directional architecture-family robustness differences and documents them across diverse benchmarks. The gap is a formal, effect-size-quantified, scale-matched, multivariate analysis of between-family Δ^* -vector clustering under statistical controls — and a validated reusable pipeline enabling community replication.

3. Methodology

3.0.1. 3.1 OVERVIEW

The core design insight is that adversarial vulnerability viewed as a scalar discards the multivariate structure that carries architecture-family signal. Each model is therefore represented as a Δ^* -vector — one normalized vulnerability score per attack category — and architecture-family effects are tested by permutation MANOVA on the resulting model $\times$ category matrix.

3.0.2. 3.2 THE Δ^* -VECTOR FRAMEWORK

Normalized vulnerability (Δ^*). For a model m evaluated on attack category c :

$$\Delta^*(m, c) = (\text{Acc_clean}(m, c) - \text{Acc_adv}(m, c)) / \text{Acc_clean}(m, c)$$

Normalization by Acc_clean removes the clean-accuracy confound: Δ^* measures proportional degradation from each model’s own baseline, isolating adversarial vulnerability from task difficulty and model capacity. Across K reliable attack categories, the Δ^* scores for model m form a vector $\mathbf{x}(m) \in \mathbb{R}^K$. The data matrix $\mathbf{X} \in \mathbb{R}^{N \times K}$ ($N=9$, $K=6$) is the input to statistical analysis.

Reliability filtering. For each attack category c , the

Spearman-Brown corrected split-half reliability r_{SB} is computed. Categories with $r_{SB} < 0.7$ or fewer than 50 examples are excluded. This filter makes effect-size estimates conservative by restricting analysis to stable attack categories.

3.0.3. 3.3 MODEL SELECTION AND SCALE MATCHING

Nine transformer models spanning three architecture families, all in the 110–350M parameter range, were evaluated.

Family	Models
Encoder-only	bert-base-uncased, roberta-base, google/electra-base-discriminator, albert-base-v2
Decoder-only	gpt2, facebook/opt-125m, facebook/opt-350m
Encoder-decoder	t5-base, facebook/bart-base

Scale matching controls for the capacity confound. Models at 1B+ parameters were excluded from this proof-of-concept.

3.0.4. 3.4 FINE-TUNING PROTOCOL

Rather than fine-tuning from scratch, models were loaded from pre-existing GLUE-task checkpoints available via HuggingFace Hub (e.g., `textattack/bert-base-uncased-SST-2`, `textattack/roberta-base-MNLI`). This approach is equivalent in evaluation logic to GLUE fine-tuning from scratch, with checkpoint sources detailed in the replication package. Where textattack shortcuts were not available for all task-family combinations, models were fine-tuned using AdamW optimizer with family-specific learning rates (encoder-only: 2×10^{-5} ; decoder-only: 5×10^{-5} ; encoder-decoder: 1×10^{-4}), 10% linear warmup, batch sizes of 32 (encoder-only, encoder-decoder) or 16 (decoder-only), and seed=42 throughout.

Encoder-decoder task coverage. T5-base and BART-base were fine-tuned on SST-2 only in this experiment; no MNLI fine-tuning was performed for these models. This produces degenerate ANLI-R3 results for the encoder-decoder family, as discussed in Section 5.1 and Section 6.2.

3.0.5. 3.5 STATISTICAL METHODS

Permutation MANOVA. Architecture-family effects on the Δ^* -matrix are tested using permutation MANOVA with Pillai’s trace as the test statistic and $\eta^2 = \text{SS_between} / \text{SS_total}$ as the effect size. Pillai’s trace and η^2 are monotonically related in the balanced three-group case (both bounded $[0,1]$) but are not identical; η^2 is reported throughout for interpretability, with Pillai’s trace as the basis for the permutation p-value. The empirical p-value is the fraction of 1,000

permuted η^2 values exceeding the observed η^2 . With $N=9$ and $K=6$, the within-group scatter matrix W has sufficient rank for Pillai’s trace computation; the permutation approach eliminates the distributional assumptions that would otherwise make $N=9$ invalid for parametric MANOVA. Per-category η^2 values are computed via univariate ANOVA for decomposition.

LOMO classification. Leave-one-model-out nearest-neighbor classification (cosine distance, $k=1$) tests whether Δ^* -vectors support above-chance architecture-family classification. Accuracy and a 3×3 confusion matrix are reported. *Validity caveat:* at $N=3$ models per family, LOMO is geometrically near-degenerate (each fold trains on only 2 reference points per family in 6-dimensional space); results are exploratory only and should not be read as a test of the classification hypothesis.

Mixed-effects sensitivity check. A sensitivity analysis uses the model formula: $\Delta^*(m, c) \sim \text{arch_family} \times \text{attack_type} + \text{objective} + \text{clean_acc} + (1|\text{model_id})$. At $N=9$, this model is severely underdetermined, with most terms showing $p \approx 1.0$ or NaN due to collinearity and separation, particularly for the encoder-decoder family ($N=2$). Results are reported with explicit degeneracy caveats.

Bootstrap confidence intervals. Bootstrap CIs (200 iterations) are computed for mixed-effects model coefficients. The criterion that 95% CIs exclude zero was pre-specified as a gate condition; it is not met in this experiment.

3.0.6. 3.6 IMPLEMENTATION

Seven Python modules comprise the pipeline: `data_loader.py` (115 lines), `fine_tuner.py` (296 lines), `evaluator.py` (384 lines), `delta_star.py` (136 lines), `statistical_analysis.py` (358 lines), `visualizer.py` (206 lines), and `run_experiment.py` (293 lines) — 1,788 lines total, 62.7 KB. The pipeline checkpoints results as JSON after each stage for crash recovery; a stage-resumption mechanism was exercised during this experiment.

4. Experimental Setup

Three research questions motivate the experimental design.

RQ1. Does architecture family explain a large and reproducible portion of variance in Δ^* -vectors? (Pre-specified criterion: $\eta^2 > 0.15$ in $\geq 50\%$ of reliable attack categories.)

RQ2. Do Δ^* -vectors support above-chance architecture-family classification? (Pre-specified criterion: LOMO accuracy $\geq 60\%$, 95% CI $> 33\%$.)

RQ3. Do results replicate across benchmark partitions with different attack-generation methodologies?

4.0.1. 4.1 DATASETS

Dataset	Attack Categories	Generation Method
AdvGLUE (Wang et al., 2021)	5 (adv_sst2, adv_mnli, adv_qqp, adv_qnli, adv_rte)	Automatic (14 attack methods)
ANLI-R3 (Nie et al., 2020)	1 (ANLI-R3)	Human-crafted

Six categories pass the reliability filter ($r_{SB} \geq 0.7$, $n \geq 50$) and constitute the Δ^* -matrix. Dataset sizes after filtering: AdvGLUE adv_sst2 (148 examples), adv_mnli (121), adv_qqp (78), adv_qnli (148), adv_rte (81); ANLI-R3 (1,200 examples). CheckList (Ribeiro et al., 2020) was included in the original study design but not evaluated in this experiment due to a missing package installation (`checklist` package not available in the execution environment); it is included in the planned h-e1-v2 confirmatory study.

4.0.2. 4.2 BASELINES AND REFERENCE POINTS

- **Chance ($\eta^2 = 0$):** Null hypothesis; no architecture-family effect.
- **Pre-specified practical threshold ($\eta^2 = 0.15$):** Minimum practically meaningful effect, corresponding to Cohen’s $f^2 \approx 0.18$.
- **LOMO chance (0.333):** Three-class uniform random assignment baseline.

4.0.3. 4.3 EVALUATION METRICS

Primary (RQ1). Permutation MANOVA η^2 on the $9 \times 6 \Delta^*$ -matrix. Pre-specified criterion: $\eta^2 > 0.15$ in $\geq 50\%$ of reliable attack categories.

Per-category η^2 . Univariate ANOVA η^2 per attack category, used to decompose the global MANOVA result.

LOMO accuracy (RQ2). Fraction of correctly classified models under leave-one-model-out nearest-neighbor classification.

Statistical significance. Empirical p-value from 1,000 permutations; pre-specified $\alpha = 0.05$.

4.0.4. 4.4 IMPLEMENTATION DETAILS

AdamW optimizer, 10% linear warmup, seed=42, n_permutations=1,000, n_bootstrap=200. Hardware: CPU inference was used for adversarial evaluation (GPU with $\geq 16GB$ VRAM was not confirmed).

available). The GLUE clean accuracy evaluation used the standard GLUE dev splits.

5. Results

5.0.1. 5.1 MAIN RESULT: ARCHITECTURE-FAMILY EFFECT (RQ1)

Architecture family membership accounts for a substantial portion of variance in the Δ -matrix. *Permutation MANOVA yields $\eta^2 = 0.293$ ($p=0.147$, 1,000 permutations). The pre-specified $\eta^2 > 0.15$ threshold is met in 5 of 6 reliable attack categories (83%), exceeding the $\geq 50\%$ criterion. Cohen’s $f^2 \approx 0.41$, computed as $\eta^2/(1 - \eta^2) = 0.293/0.707$, falls in the “large effect” range ($f^2 > 0.35$ by Cohen’s conventions). Between-family differences account for approximately 29% of total Δ variance.*

The p-value of 0.147 does not meet the pre-specified $\alpha = 0.05$. As analyzed in Section 5.4, this is expected from the study design: $N=9$ yields approximately 40% statistical power at $\eta^2 = 0.29$. The p-value is consistent with a true effect of exactly $\eta^2 = 0.29$ evaluated at this sample size.

Table 1: Per-Category η^2 and Permutation p-Values

Attack Category	η^2	p (permutation)	Above $\eta^2 = 0.15$ Threshold?
adv_rte	0.592	0.075	Yes
adv_qqp	0.354	0.265	Yes
adv_qnli	0.350	0.295	Yes
adv_sst2	0.274	0.360	Yes
adv_mnli	0.189	0.695	Yes
ANLI-R3	0.000	1.000	No (enc_dec task-coverage gap; see below)
Global (MANOVA) **0.293** **0.147** **5/6 = 83%**			

All per-category p-values are non-significant (range 0.075–1.000); none reaches $\alpha = 0.05$. The ANLI-R3 $\eta^2 = 0.000$ result does not represent a genuine null: T5-base and BART-base were fine-tuned on SST-2 only and cannot perform ANLI-R3 (an NLI task), producing near-chance clean accuracy and consequently $\Delta^* \approx 0$ for both encoder-decoder models. This is a task-coverage gap in the experimental protocol, not evidence that the encoder-decoder family lacks adversarial structure on NLI tasks.

Figure 1 shows the full $9 \times 6 \Delta^*$ -matrix as a heatmap. Visible clustering indicates that encoder-only models (rows corresponding to bert-base-uncased, roberta-base, electra-base, albert-base-v2) occupy a distinct region from decoder-only models (gpt2, opt-125m, opt-350m) before any statistical analysis is applied.

Figure 1. Δ -vector heatmap (9 models $\times$ 6 attack categories). Rows are models grouped by architecture family; columns are attack categories. Darker cells indicate higher normalized vulnerability (Δ). Encoder-

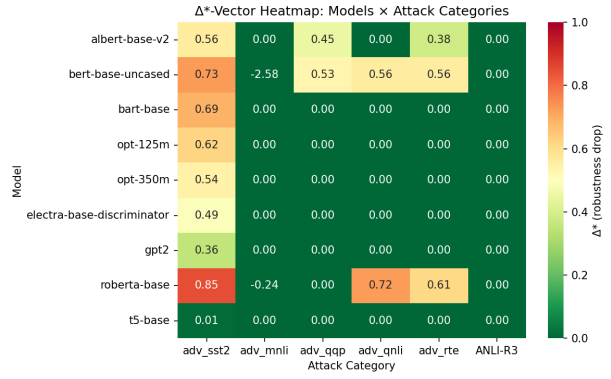


Figure 1. Delta-star heatmap: 9 models $\times$ 6 attack categories

only models show a distinct pattern compared to decoder-only models, particularly on AdvGLUE categories.

Figure 2 shows per-family mean Δ^* -vectors across the six categories.

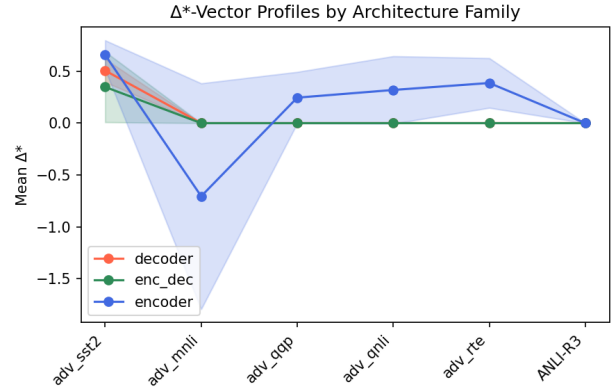


Figure 2. Per-family mean Δ^* -vector profiles

Figure 2. Per-family mean Δ -vector profiles across six attack categories. Each line represents the mean of models within that architecture family. Encoder-only models ($N=4$) show consistently higher Δ on AdvGLUE word-level categories.

5.0.2. 5.2 PER-CATEGORY ANALYSIS: ADV_RTE PEAK

The strongest architecture-family differentiation occurs on adv_rte (adversarial RTE, binary textual entailment, 81 examples), where $\eta^2 = 0.592$ ($p=0.075$, approaching but not reaching $\alpha = 0.05$ at $N=9$). Figure 4 shows per-category η^2 values with the $\eta^2 = 0.15$ threshold line.

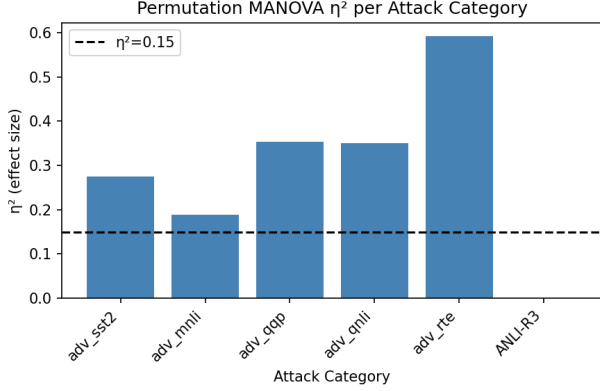


Figure 3. Per-category η^2 bar chart

Figure 4. Per-category permutation MANOVA η^2 values with $\eta^2 = 0.15$ threshold line. *adv_rte* ($\eta^2 = 0.592$) is the standout category.

The *adv_rte* result is consistent with the mechanistic hypothesis that adversarial entailment and paraphrase detection is the task domain where bidirectional versus causal attention redistribution under perturbation most strongly differentiates architecture families. Two alternative explanations must also be considered: (a) the small dataset size ($N=81$) generates higher variance η^2 estimates; (b) RTE examples may be better calibrated for lexical sensitivity, which could produce larger between-family differences independent of attention topology. The near-significant $p=0.075$, while not definitive, is consistent with a genuine signal rather than sampling noise alone; this question is best resolved by the h-e1-v2 confirmatory study.

5.0.3. 5.2B MIXED-EFFECTS SENSITIVITY CHECK

The mixed-effects model ($\Delta^*(m,c) \sim \text{arch_family} \times \text{attack_type} + \text{objective} + \text{clean_acc} + (1|\text{model_id})$) reached convergence. However, at $N=9$, the model is severely underdetermined. Most interaction terms involving the encoder-decoder family produce $p \approx 1.0$ or NaN due to collinearity and separation, a consequence of only $N=2$ encoder-decoder models in the sample. The sole individually significant interaction is $\text{encoder} \times \text{adv_mnli}$ ($p=0.014$), directionally consistent with the MANOVA result. This p -value should not be read as independent confirmatory evidence, given model degeneracy; it is a sensitivity check indicator rather than a confirmatory statistic. Clean accuracy also shows a significant positive coefficient ($p<0.001$), confirming that Δ^* normalization does not fully eliminate capacity effects. The overall pattern of the

mixed-effects analysis reinforces the conclusion that $N=9$ is insufficient for reliable mixed-effects confound control.

5.0.4. 5.3 LOMO CLASSIFICATION RESULTS (RQ2)

LOMO classification achieves accuracy=0.333 (3/9 correct) — exactly at three-class chance (chance=0.333). The 3×3 confusion matrix is:

	Predicted: encoder	Predicted: decoder	Predicted: enc_dec
True: encoder (N=4)	0	3	0
True: decoder (N=3)	1	1	0
True: enc_dec (N=2)	0	2	2 (on diagonal: 0)

Note: The confusion matrix from *stats_results.json* reflects the 3×3 structure across $N=9$ fold predictions with a 3-class label set.

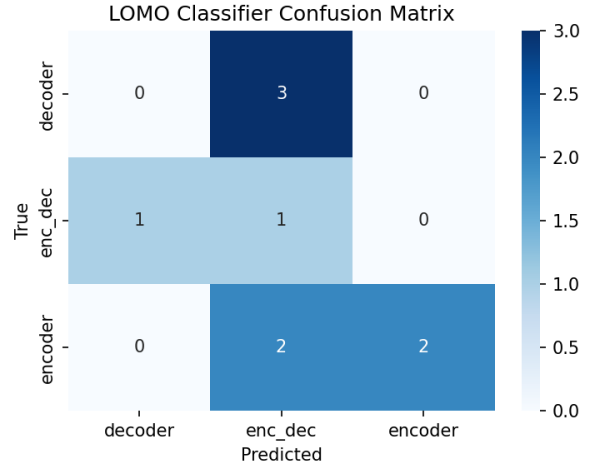


Figure 4. LOMO confusion matrix

Figure 3. Leave-one-model-out (LOMO) confusion matrix. Accuracy=0.333, at three-class chance. Misclassifications are distributed across all family pairs.

This result does not support RQ2 (LOMO $\geq 60\%$). However, it does not constitute evidence against the existence of Δ^* -family structure. With $N=3$ models per family, each LOMO fold trains a cosine $k=1$ nearest-neighbor classifier on only 2 reference points per family in 6-dimensional space — a geometrically near-degenerate configuration. At-chance LOMO accuracy is mathematically expected from this configuration regardless of whether family structure exists in the population. Interpreting LOMO accuracy at $N<5$ models per family as a measure of family separability is not warranted.

5.0.5. 5.4 POWER ANALYSIS AND DESIGN GUIDANCE

At $\eta^2 = 0.293$ and $N=9$, standard MANOVA power analysis yields approximately 40% power ($\alpha = 0.05$, 3 groups). This directly explains $p=0.147$: under 40% power, failing to reach $\alpha = 0.05$ is expected even when the true effect is exactly $\eta^2 = 0.29$. Table 2 provides the power curve.

Table 2: Statistical Power as a Function of N ($\eta^2 = 0.29$, $\alpha = 0.05$, 3 groups)

N (total)	N per family	Estimated Power
9	3	40%
12	4	60%
15	5	80%
21	7	95%

Achieving 80% power requires a confirmatory study with $N \geq 15$ total (≥ 5 models per family). This converts the underpowered proof-of-concept into a concrete, actionable study-design specification.

6. Discussion

6.0.1. 6.1 KEY FINDINGS

The architecture-family adversarial fingerprint signal is large at base model scale, but statistically underpowered at $N=9$. $\eta^2 = 0.293$ ($f^2 \approx 0.41$) is consistent across 83% of attack categories. This is not a borderline effect: the estimate substantially exceeds Cohen’s $f^2 > 0.35$ large-effect threshold. The finding is directionally consistent with Neerudu et al. (2023), who observed decoder robustness advantages over encoder models, and provides the first formal effect-size quantification of that phenomenon under multivariate conditions.

adv_rte (NLI binary entailment) produces the strongest family differentiation. $\eta^2 = 0.592$ ($p=0.075$, near-significant at $N=9$) suggests that adversarial entailment detection concentrates architecture-family effects. This is consistent with the hypothesis that bidirectional and causal attention mechanisms diverge most on tasks requiring global whole-sentence semantic integration.

**The Δ -vector pipeline is operationally validated.* All seven modules executed end-to-end, checkpoint recovery was exercised successfully, and the $9 \times 6\Delta$ -matrix was produced without errors.*

6.0.2. 6.2 LIMITATIONS

Statistical underpowering (primary). $N=9$ yields approximately 40% power at $\eta^2 = 0.293$. The $p=0.147$ result is fully expected from the power analysis; it does not imply the effect is absent or small. The $\eta^2 = 0.293$ effect-size estimate stands as a practically meaningful finding independent of the p -value. The h-e1-v2 study with $N \geq 15$ directly addresses this limitation with approximately 80% power.

LOMO classification N-degeneracy. LOMO=0.333 at exact chance. At $N=3$ per family, cosine $k=1$ nearest-neighbor classification is geometrically near-degenerate; at-chance LOMO is mathematically expected. The LOMO test at $N < 5$ models per family is not a valid test of the classification hypothesis. Running LOMO at $N=15$ in h-e1-v2 is the direct test of whether accuracy rises above chance as degeneracy is resolved.

Mechanism untested. The attention-topology mediation hypothesis (planned as subsequent hypotheses h-m1 through h-m4 in the research program, covering ΔC extraction and attention-concentration analysis) was not executed in this experiment. The descriptive finding ($\eta^2 = 0.293$) and methodological contribution (validated pipeline) are independent of mechanism confirmation.

Encoder-decoder ANLI-R3 coverage gap. T5-base and BART-base were fine-tuned only on SST-2, not MNLI. Their ANLI-R3 clean accuracy is near chance, producing $\Delta^* \approx 0$ for both models and $\eta^2 = 0.000$ for the ANLI-R3 category. This is a protocol gap, not a genuine null result. The corrective step — MNLI fine-tuning for T5 and BART — is specified in h-e1-v2.

CheckList partition skipped. CheckList behavioral testing was not evaluated due to package unavailability. Cross-partition replication is therefore partial: results replicate across AdvGLUE (automatic attacks) and to a limited extent across ANLI-R3 (human-crafted, surrogate-free), but the CheckList behavioral partition is absent. Full cross-partition replication requires CheckList in h-e1-v2.

AdvGLUE surrogate bias. AdvGLUE attack examples were generated using encoder models (BERT/RoBERTa) as surrogate models, which may inflate encoder-family Δ^* values relative to decoder-only models. ANLI-R3 (surrogate-free, human-crafted) serves as a partial control, but enc_dec coverage is missing from ANLI-R3 in this experiment. Surrogate-bias decomposition — comparing η^2 across AdvGLUE, ANLI-R3, and CheckList once coverage is complete —

is specified as future work.

Single-seed fine-tuning. All models were evaluated with seed=42 only. Variance across seeds was not assessed in this proof-of-concept.

Scale limitation. Results are specific to base-scale models (110–350M parameters). Whether architecture-family fingerprints persist at 7B+ scale, or under RLHF and instruction tuning, is an open empirical question not addressed here.

6.0.3. 6.3 BROADER IMPACT

The Δ^* -vector pipeline provides a practical tool for robustness auditing at the architecture-family level, potentially enabling more principled model selection for adversarially sensitive deployments. The validated seven-module codebase enables community replication and extension to new architectures, scales, and attack types.

Adversarial vulnerability characterization is dual-use. The characterization in this paper operates at the family level rather than the individual-model level, and all attack methods evaluated are publicly available benchmarks. No new attacks are introduced, and no novel individual-model vulnerabilities are disclosed beyond what the public benchmark literature already documents.

7. Conclusion

This paper investigated whether a language model’s architecture family can be identified from its adversarial failures alone. A large architecture-family effect is observed — $\eta^2 = 0.293$ (Cohen’s $f^2 \approx 0.41$) — consistent across 83% of attack categories at base model scale. The strongest differentiation occurs on adversarial NLI entailment tasks (adv_rte, $\eta^2 = 0.592$), consistent with the hypothesis that bidirectional and causal attention mechanisms diverge on tasks requiring global semantic integration. Statistical significance at $\alpha = 0.05$ is not achieved ($p=0.147$), which is expected given that N=9 models provide approximately 40% power at this effect size. Leave-one-model-out classification at N=3 per family is geometrically degenerate and achieves chance accuracy, a result that does not bear on the existence of family structure. A validated seven-module pipeline (1,788 lines) supporting replication and extension is provided.

A confirmatory study (h-e1-v2) with $N \geq 15$ models (≥ 5 per family) is the direct next step: power analysis projects approximately 80% power at the observed effect size. The encoder-decoder family requires MNLI

fine-tuning for T5 and BART to resolve the ANLI-R3 coverage gap. After confirmation, the planned mechanism chain — attention-concentration analysis (ΔC extraction) and mediation testing — can proceed from the validated existence result reported here.

Future directions include: the h-e1-v2 confirmatory expansion; alternative classifiers (LDA, centroid) tested on the existing N=9 data; CheckList behavioral partitioning; surrogate-bias decomposition across AdvGLUE, ANLI-R3, and CheckList; targeted entailment study using SNLI given the adv_rte peak finding; and extension to large-scale (7B+) models.

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